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Shedding light on correlation mechanism between the keyhole/melt pool behaviors and photoelectric radiation information during laser welding process

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Abstract

Laser welding technology offers a novel technical approach for the electrical connection of current collector foils in individual electrodes within lithium-ion battery stacks. However, the industrial application of laser beam welding for internal battery connections still faces numerous limitations. Since the quality of welded joints is influenced by multiple process parameters, real-time monitoring of weld quality is particularly crucial. Photodiode sensors, as a widely used and cost-effective monitoring tool in laser beam welding, have not yet been fully explored for their potential in internal battery connections. This study focuses on the correlation mechanism between

Outline

keyhole/melt pool dynamic behavior and photoelectric radiation information during laser welding, aiming to provide a theoretical foundation for optimizing process parameters and enabling intelligent welding monitoring. Through high-speed imaging technology and keyhole/melt pool extraction algorithms, the intrinsic relationship between optical radiation signals and the macroscopic motion and periodic variations of the melt pool was systematically investigated. Based on dynamic thermal map analysis of the melt pool, a joint analysis of photoelectric signals and transient characteristics of the keyhole/melt pool was achieved, and statistical methods were employed to evaluate the correlation between photoelectric signal features and weld formation quality. The results indicate that, from a frequency domain perspective, the low-frequency fluctuations (0–2000Hz) of the melt pool are significantly correlated with the low-frequency components of the optical radiation signals. By summarizing the variation patterns of unstable melt pool signals, the dynamic evolution process of aluminum melt pool ejection was revealed. From a time-domain perspective, the instantaneous fluctuations of photoelectric signals are directly related to melt oscillation and keyhole dynamic behavior. Furthermore, this study explored the quantitative relationship between process parameters, optical radiation signals, and weld formation quality, confirming that optical radiation signals can effectively characterize welding quality. The findings provide a significant foundation for achieving efficient and precise laser-based internal battery connections through photoelectric monitoring methods.

Keywords

Aluminum alloys laser welding; Battery connections; Photoelectric signal; Process monitoring; Weld quality

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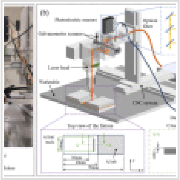

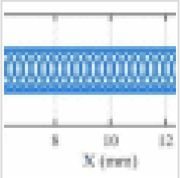

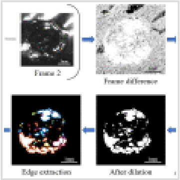

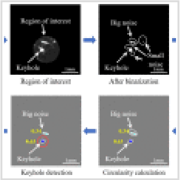

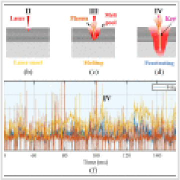

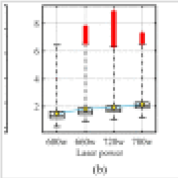

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Tables (2)

1. Introduction

Laser welding technology has been widely applied in industrial manufacturing, including the production of large-format lithium-ion batteries (LIBs), due to its high energy density, high production efficiency, ease of automation, and minimal thermal deformation [1,2]. In the production of LIBs, the process chain typically involves electrode fabrication, cell assembly, and cell finalization [3,4]. A critical step in the cell assembly is cell-internal contacting, where the uncoated regions of the current collector foils—copper foils for anodes (6–12 μm) and aluminum foils for cathodes (10–15 μm)—are joined with an arrester tab [5]. In the cell-level welding procedure, a substantial number of current collectors are bonded to the Al tab and Cu tab, which function as the cathode and anode leads, respectively. The quantity of current collectors is contingent upon the battery's required capacity [6]. However, these thin foils are highly susceptible to mechanical stresses, often leading to wrinkling or rupture during manufacturing [7]. While ultrasonic metal welding (USMW) remains the state-of-the-art process for such contacts, its requirement for dual-sided accessibility limits its format flexibility [8]. In contrast, laser beam welding introduces a contactless approach that not only circumvents these limitations but also opens new possibilities for battery cell design by enabling innovative geometries and improving volume utilization [9]. As a result, the volumetric energy density of LIBs can be enhanced—an essential factor for extending the operating range of electric vehicles [10].

Recent studies have demonstrated significant advancements in laser welding for multi-layer foil applications in battery manufacturing. Kamat et al. employed a 515-nm green light laser to weld 30 layers of 9-μm-thick copper foils with two 0.2-mm-thick copper sheets [11]. They investigated the weld seam properties under various combinations of laser power and welding speed and established a laser welding model for multi − foil stacking. Wan et al. achieved robust welding of current collector pure copper foil stacks using a blue − wavelength laser [12]. Additionally, Olowinsky et al. systematically explored the welding of metal foils using lasers of different wavelengths (blue, green, and infrared). They identified suitable process parameters for copper foils and indicated that inserting additional metal sheets could improve the issues of pores and cracks in aluminum foil welding [13]. In parallel, recent research on oscillating laser-arc hybrid welding (LAHW) has emphasized the importance of melt pool flow, keyhole stability in improving weld quality and suppressing defects across various materials and joint configurations[[14], [15], [16]]. These findings provide valuable insight into the physical mechanisms governing energy coupling and process stability. However, most of these studies have been conducted on thick plates or titanium alloys, with limited attention to photoelectric signal-based monitoring in thin, multi-layer foil structures. This highlights the need for further investigation into signal-behavior correlations under such constrained conditions.

In summary, the current application of laser welding in multi-layer foil welding has demonstrated significant advantages, particularly in terms of joint design and process parameter optimization. Despite these promising benefits, the complex nature of laser–metal interactions can negatively impact weld quality and stability, potentially leading to defects such as shallow or deep penetration, high levels of porosity, and metal spatters [[17], [18], [19]]. In the specific context of laser welding multi-layer aluminum foils within large-format LIBs, achieving reliable online monitoring and process control is of paramount importance to ensure optimal product quality [20]. Advanced optical monitoring methods—particularly those based on photoelectric sensors—offer a powerful solution by enabling real-time capture and analysis of the optical radiation signals produced during welding [21].

Photodiode-based photoelectric monitoring has emerged as a promising technique for laser welding due to its ability to capture multi-waveband optical radiation signals that are closely linked to welding process dynamics and quality [22]. According to Eriksson et al., photodiode systems in laser welding are typically used to measure emissions from welding fumes and vapors, thermal radiation, and the laser beam's back reflection, with distinct photodiodes independently detecting various bandwidths; signals in the visible (VIS) and near-infrared (NIR) ranges, for example, can correlate with the thermal state of the molten metal, thus aiding in defect detection and process understanding [23].

Chianese et al. further showed that in battery cell applications, the signals captured in the visible and NIR spectra are significantly correlated with laser power and welding depth, and that leveraging machine learning techniques on these signals can facilitate the automatic detection of process defects—though challenges remain when large part-to-part gaps introduce high fluctuations [24].

In parallel, research on deep penetration welding has revealed that the optical radiation signals, comprising ultraviolet, visible (300–700 nm), laser reflection (1064–1070 nm), and infrared (1100–1600 nm) wavebands, carry extensive information regarding welding quality. Studies by Bono et al [25]. have linked fluctuations in photodiode signals (600–850 nm) to anomalies in laser power settings, joint preparation, and fit-up. While frequency analysis, as noted by Schmidt et al. and Zhang et al., can discern distinct oscillation regions corresponding to melt pool and keyhole dynamics [26,27]. Collectively, these findings highlight that photodiode-based monitoring not only enhances our understanding of the laser-metal interaction but also provides a robust basis for real-time process control and defect identification in laser welding, although further research is required to extend these applications to scenarios with more complex welding configurations, such as those involving multiple, thinner joining partners within battery cells. In the context of LiB manufacturing, the quality of laser welding for current collector foils directly affects the safety, reliability, and electrochemical performance of battery cells. However, existing industrial monitoring methods, such as threshold-based signal discrimination or offline inspection, often struggle to detect subtle changes in the melt pool and keyhole behavior during welding, leading to undetected defects.

In our previous research, a multi-source monitoring platform based on photoelectric sensors was established. The relationship between photoelectric signals and plasma as well as penetration depth was systematically explored, and a novel framework for identifying the penetration status was developed [28]. Building on this foundation, the current study is dedicated to further exploring the correlation between keyhole/melt pool behavior and photoelectric signals. The dynamic interplay between keyhole formation, melt pool evolution, and the corresponding multi-waveband optical emissions is analyzed to unravel the underlying physics governing weld quality and stability. The insights derived from this investigation are anticipated to enhance the fundamental understanding of laser welding processes and provide a theoretical basis for developing advanced monitoring systems for industrial laser welding, ultimately contributing to more intelligent and efficient manufacturing processes for LiB current collectors.

2. Materials and methods

2.1. Experimental setup

As shown in Fig. 1, the experimental monitoring platform consists of a fiber laser welding system, a high-speed photography system, and a photoelectric signal acquisition system. The fiber laser welding system is composed of a continuous laser, a welding head, and a CNC control system. The continuous laser source (EVERFOTON FFRC-6000C) has a maximum output power of 6000W, with a central wavelength of 1080±5nm and an output fiber core diameter of 100μm. The CNC control system is used to adjust the motion parameters during the welding process. High-speed photography is employed to collect information on the melt pool and keyhole. The photoelectric sensing system consists of three sensors: the T sensor, the R sensor, and the P sensor. The T sensor includes a photodetector (PDA20CS2, THORLABS) with a detection range of 800–1700nm and a bandpass filter (1100–2100nm) for detecting near-infrared radiation. The R sensor comprises a photodetector (PDA100A2, THORLABS) with a detection range of 320–1100nm and a narrowband filter (1075±2 nm, FWHM 32–33nm) for detecting the laser reflected by the material. The P sensor is composed of a PDA100A2 photodetector and a bandpass filter (300–700nm) for detecting visible light radiation. All three sensors are connected to a data acquisition card, with the sampling frequency set at 40kHz. The high-speed camera used is the Phantom VEO 710L, equipped with a macro lens (AF Micro-Nikkor ED 200mm f/4D IF, Nikon). The lens aperture is set to F5.6, with a resolution of 1024×512, a sampling rate of 10,000 fps, an exposure time of 20μs, and an exposure index (EI) set to 6400. An 810nm long-pass filter and an active light source (CAVILUX HF, CAVITAR) are attached to the lens to reduce interference from high-intensity radiation and optimize imaging quality.

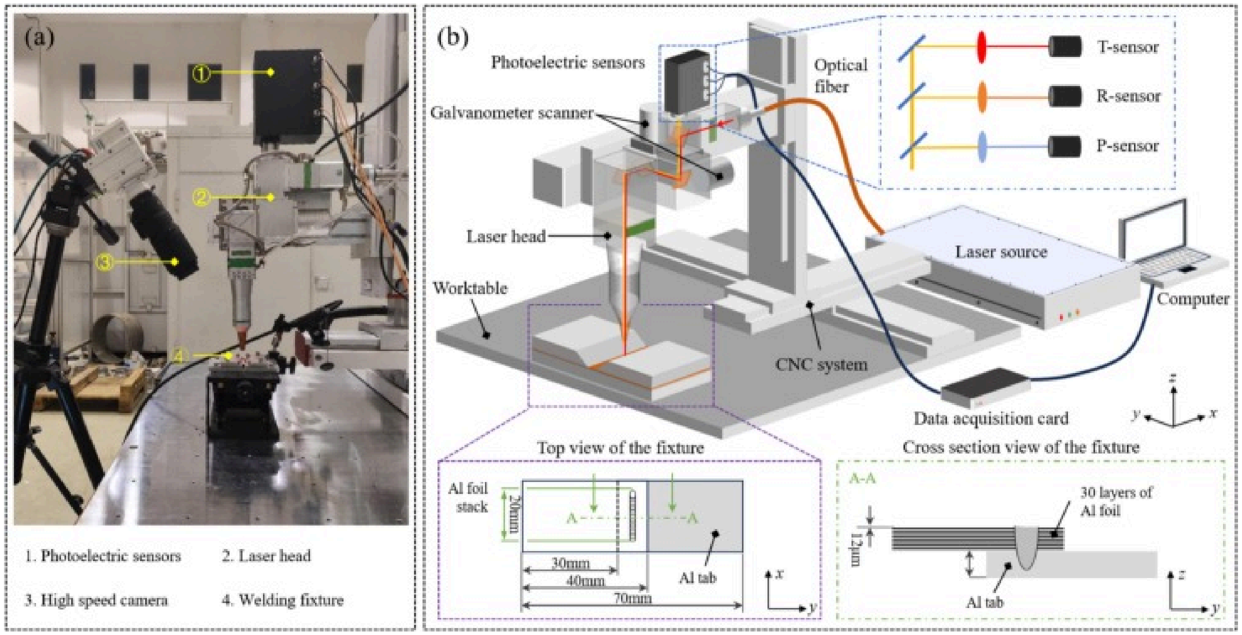

Fig. 1. View of the experimental platform.

To ensure a gap-free welding configuration for the foil stack, a fixture as shown in Fig. 1(b) was employed. For the production of large-sized lithium-ion batteries, an application-oriented welding task was evaluated, which involved connecting 30 layers of 12 μm battery-grade aluminum foil to a 0.3 mm thick arrester tab. The dimensions and materials of the test samples are presented in Table. 1. The welding sample was a rectangular piece measuring 70 mm × 25 mm, with an overlap of 10 mm between the foil stack and the joint. Each weld seam was 20 mm long, located at the center of the overlapping area, as depicted in Fig. 1(b). Laser power directly determines the amount of energy input and is a key factor affecting the melting and evaporation behavior of the material. Therefore, the welding experiment used laser power as a variable to investigate the correlation between different melt pool keyhole modes and photoelectric signals, as well as the impact on welding quality. Table. 2 shows the detailed welding parameters. The welding trajectory was a spiral line, as illustrated in Fig. 2.

Table 1. Detailed welding parameters and corresponding welding quality.

Case	Laser power p (W)	Welding speed v (mm/s)	Wobble mode	Wobble radius r_w(mm)	Wobble frequency f_w(Hz)
1	600	12	circle	0.8	40
2	660				
3	720				
4	780				

Table 2. Overview of the used welding geometries and the used materials.

Material	Dimension (mm)	Layers	Weld seam length (mm)
1060 Al foil	40.0×25.0×0.012	30	20.0
1060 Al tab	40.0×25.0×0.3	1	

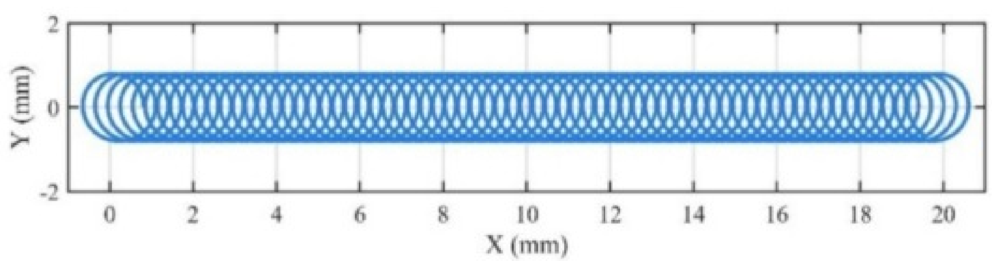

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Fig. 2. Schematic diagram of welding trajectory.

2.2. Image analysis

Fig. 3 depicts the detailed procedure for extracting the melt pool contour from high-speed photography images using the frame difference method. Initially, the frame difference is computed from two consecutive high-speed photography images. This process highlights the dynamic features within the melt pool by identifying dynamic changes in the image through the absolute difference of pixel intensity values. Given that molten metal exhibits a high degree of transparency, a single static image is insufficient for accurately extracting the melt pool contour. Therefore, the dynamic differences between frames are leveraged to enhance contour information. Subsequently, the generated difference image exhibits considerable noise and dynamic regions, which necessitate further processing for accurate contour extraction. A threshold-based segmentation technique is employed to generate an initial motion mask, which identifies regions exhibiting significant changes. To mitigate the impact of noise, a morphological filter is applied to the motion mask, effectively eliminating isolated pixels and small noise regions. Subsequently, a dilation operation is performed on the mask to merge fragmented regions that may belong to the same melt pool area, thereby improving the continuity of the dynamic region. Edge detection is then conducted to identify the preliminary contour of the melt pool, and the final contour is determined through convex hull calculation, ensuring the completeness and closure of the contour. This process significantly enhances the accuracy of the melt pool contour extraction by focusing on dynamic regions, thereby providing a robust basis for subsequent analysis.

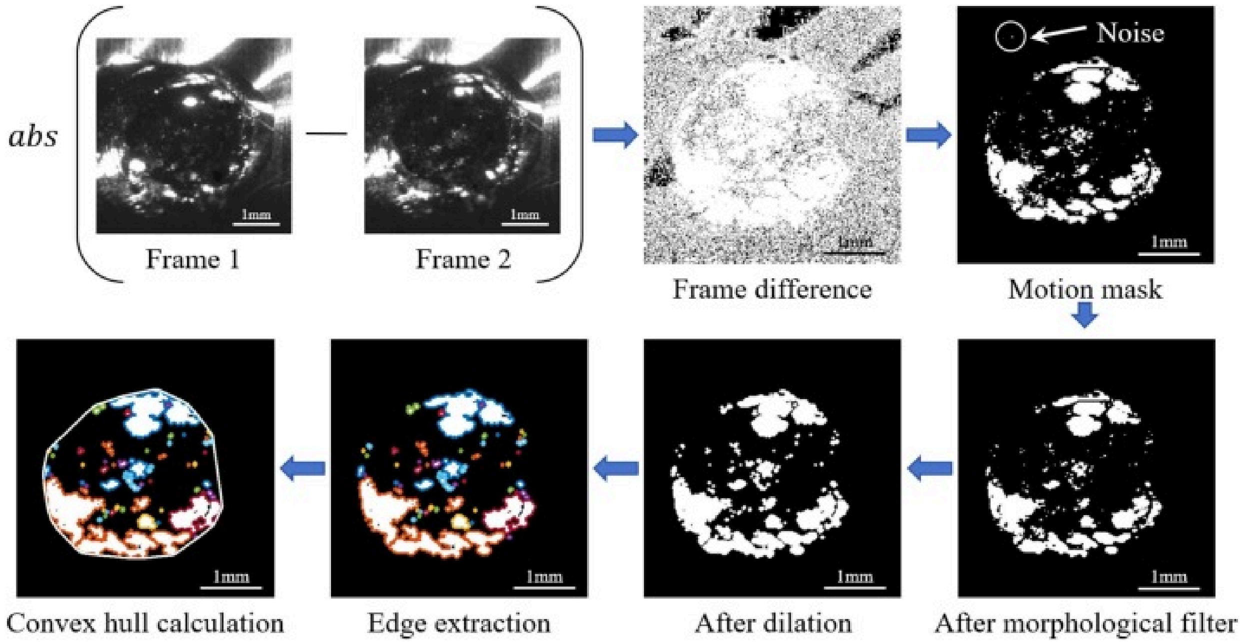

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Fig. 3. Melt pool extraction process based on frame difference and convex hull calculation.

To further extract smaller keyhole features from the melt pool images, a keyhole extraction algorithm based on trajectory tracking and circularity calculation was developed, with the specific extraction process depicted in Fig. 4. This algorithm initially selects the Region of Interest (ROI) containing the welding area from the original high-speed photography images, thereby isolating keyhole features and minimizing background interference. The ROI selection is determined by the movement trajectory of the laser beam, and the equation describing this trajectory is presented as follows:

$$\omega = 2 \cdot \pi \cdot f_w \quad (1)$$

$$\mathrm{x}(t) = r_w \cos(\omega t) + v \cdot t \quad (2)$$

$$\mathrm{y}(t) = r_w \sin(\omega t) \quad (3)$$

In the equation, ω represents the angular velocity, f_w denotes the oscillation frequency, r_w is the oscillation radius, and v signifies the welding speed. Subsequently, the ROI image undergoes binarization to initially isolate the keyhole from the background while preserving some noise information. Then, a two-step process involving the removal of small and large noise regions is applied to eliminate further interference, leaving only the keyhole features. In the denoised image, the edges of the detected keyhole features are extracted, and their morphological characteristics are quantified by calculating the circularity value. This value is used to determine whether the detected features match the expected morphological characteristics of the keyhole, thus enabling precise extraction of the keyhole. The formula for calculating the circularity value is as follows:

$$C = 4\pi A/P^2 \tag{4}$$

In the formula, A denotes the area of the object, and P represents the perimeter of the object. This method, by combining trajectory tracking with morphological feature analysis, effectively enhances the accuracy and robustness of keyhole detection.

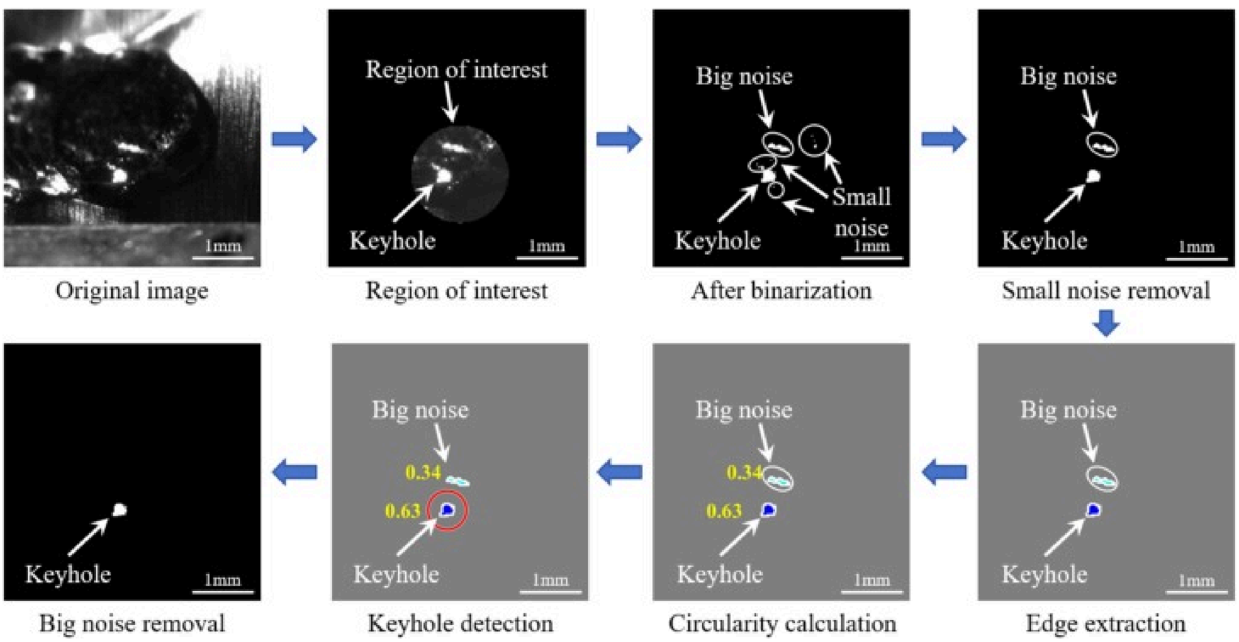

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Fig. 4. Keyhole extraction algorithm based on welding trajectory tracking and morphological calculations.

3. Results and discussion

3.1. The generation and evolution mechanism of optical radiation signals

The laser welding process is a complex thermomechanical coupling process, which can be divided into five stages based on the distinctive features of photoelectric signals: preparation (I), laser activation (II), melting (III), penetration (IV), and solidification (V). Fig. 5(a–e) depict the schematic diagrams of these five stages, with detailed descriptions of the events occurring in each stage provided in the corresponding figure captions. Under continuous laser irradiation and the influence of heat accumulation, the behavior of the melt pool and keyhole undergoes significant changes. Analyzing these dynamic changes provides insights into the formation and evolution mechanisms of photoelectric signals. Fig. 5f presents the optical radiation data for a typical welding process. From the first stage to the third stage, the material surface begins to absorb thermal energy, resulting in an increase in surface temperature and a corresponding enhancement of reflected light and infrared radiation. This stage marks the initial energy input, with heat accumulating to facilitate the subsequent melting and penetration stages. As the process enters the fourth stage, heat is conducted into the material's interior, deepening the melt pool and enabling gradual penetration through the material. A high-temperature, high-pressure plasma or vapor channel, known as the keyhole, forms, leading to a sharp increase in infrared radiation (T-Signal) and a significant enhancement of reflected light (R-Signal). In the fifth stage, laser irradiation ceases, and the P-Signal and R-Signal diminish accordingly. The melt pool cools and solidifies, forming the final weld. As cooling progresses, T-Signal decreases significantly, indicating reduced heat release and the near completion of the welding process. At this point, the temperature and strain fields of the material can be considered quasi-steady, resulting in more stable and orderly optical radiation signals across the three spectral bands.

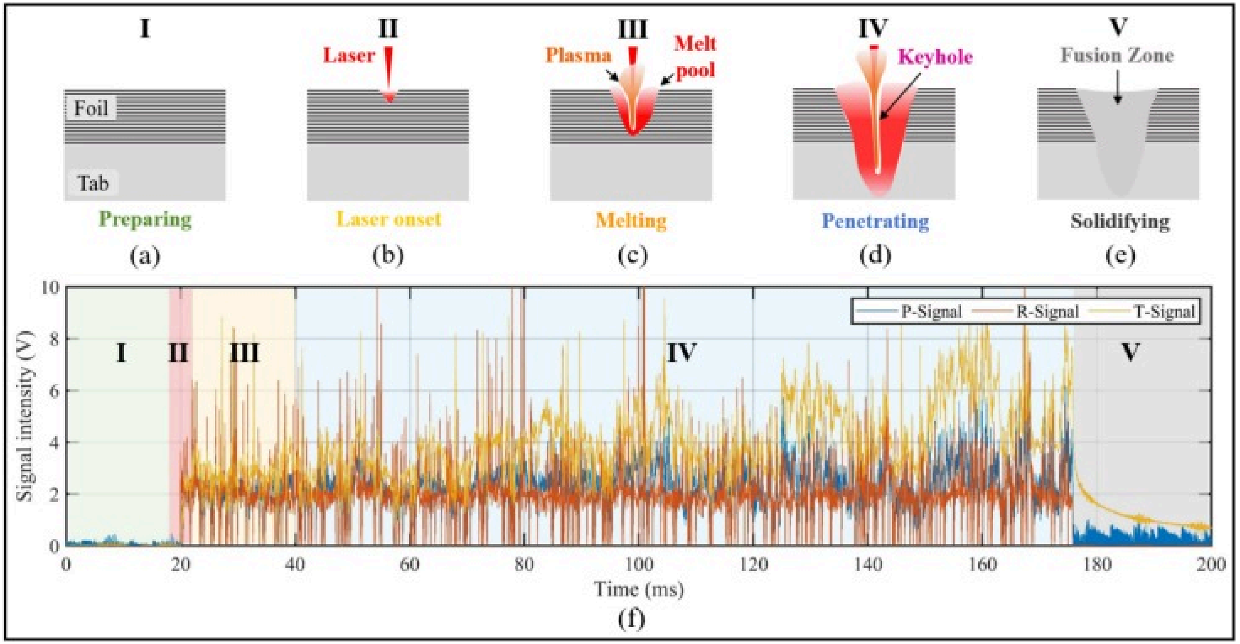

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Fig. 5. Schematic of the laser welding stage classification and typical photoelectric signals for each stage. (a) Stage I (Preparation): Laser off, no energy input to the workpiece. (b) Stage II (Activation): Laser on, beam contacts the material surface, initial heat absorption. (c) Stage III (Melting): Material melts, shallow pool forms with continued laser energy. (d) Stage IV (Penetration): Heat conducts inward, pool deepens, material penetrated. (e) Stage V (Solidification): Welding complete, workpiece cools or undergoes post-treatment. (f) Three-band optical radiation signals for each stage.

Fig. 6 presents the optical radiation signal data for different laser power levels. As the laser power increases, the intensity of visible light and near-infrared signals correspondingly increases. The reflected light signals exhibit a broad intensity range and a higher number of outliers, indicating greater variability. To further elucidate the causes of these changes in optical radiation signals, it is necessary to analyze the high-speed photography images of the melt pool and keyhole.

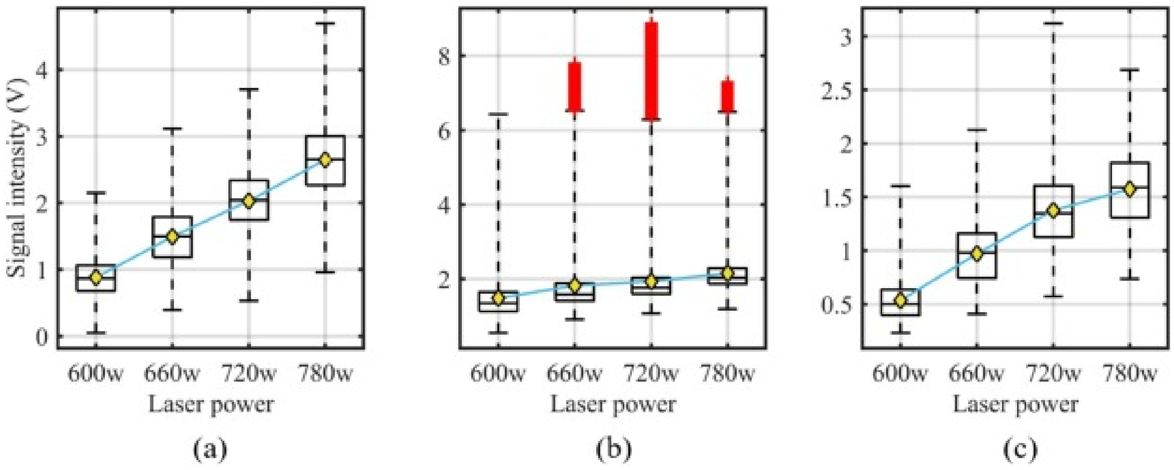

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Fig. 6. Box plot of optical radiation signals during laser welding at various laser powers. (a) P-Signal. (b) R-Signal. (c) T-Signal.

The characteristics of the melt pool are directly related to weld quality and the formation of defects, with a stable melt pool being indicative of high-quality welds. Fig. 7 presents the evolution of melt pool features captured by a high-speed camera under different parameters, including four laser power levels: 600W, 660W, 720W, and 780W. Fig. 7(a1–d5) illustrate the evolution of melt pool characteristics at these laser powers. At 600W, the line energy (*EL* = laser power / welding speed) is relatively low, resulting in significantly less material being melted compared to higher power levels. Consequently, this condition produces the smallest weld width. As reported by Wang et al., the cumulative expansion of molten metal primarily enhances heat transfer efficiency at the surface of the melt pool, resulting in a wider weld surface[29]. When the laser power is set to 660W, the increased weld energy leads to a significant expansion of the melt pool area. Under these conditions, the melt pool moves smoothly forward under the influence of the laser beam, with no splashing observed, thereby achieving high-quality melting of the aluminum foil stack (see Fig. 7(b3)). The welding trajectory exhibits well-defined geometric boundaries, with melt pool stability maintained throughout the 1.3–5 ms interval. However, at power levels of 720 W and 780 W, distinct fluctuations occur in the melt pool (Fig. 7(c1–d5)). The intensified internal flow drives curvature formation, accompanied by undulating features (peaks and valleys) as illustrated in Fig. 7(c3). Unstable fluctuations further induce liquid metal rupture or bursting (Fig. 7(d2–d4)), while splashing becomes prevalent in the melt pool (Fig. 7(c4; d5)). This paper categorizes this as an extremely unstable state.

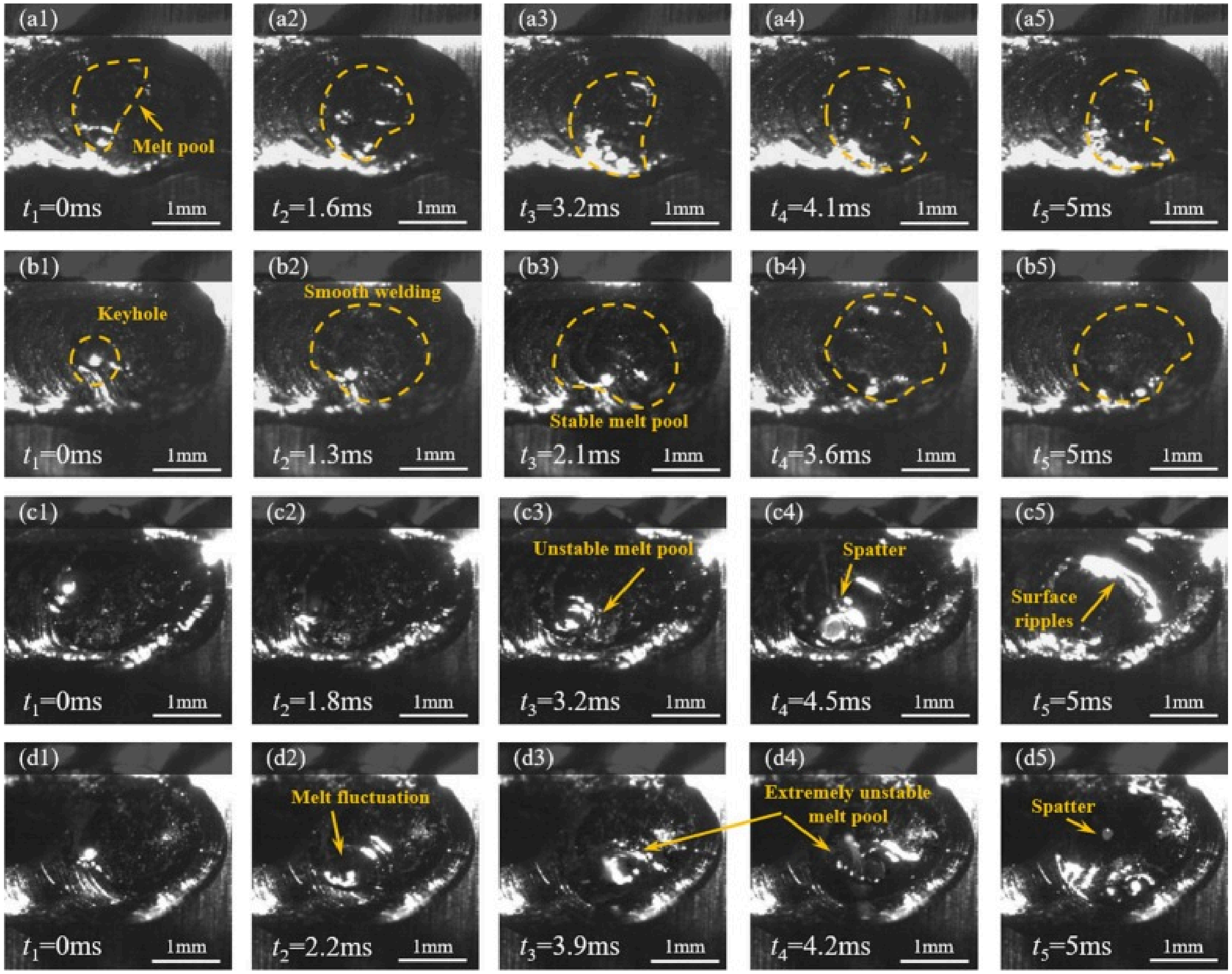

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Fig. 7. Evolutionary behavior of the melt pool at different powers. (a) Laser power=600w. (b) Laser power=660w. (c) Laser power=720w. (d) Laser power=780w.

Fig. 8 illustrates the variations in melt pool area and circularity with respect to laser power. The experimental results indicate a positive correlation between melt pool area and laser power (600–780W), accompanied by amplified fluctuation amplitudes at higher power levels. Specifically, at low power levels (600W, 660W), the melt pool area is smaller and the fluctuation is less significant; whereas at high power levels (720W, 780W), the melt pool area enlarges and the fluctuation becomes more pronounced. The circularity analysis reveals that the circularity fluctuates more at low power, while it tends to stabilize and approaches 0.9 at high power levels (660W, 720W). Although the circularity slightly decreases at 780W, it remains at a relatively high level. These observations demonstrate that laser power exceeding 660W promotes both melt pool expansion and geometric regularization, likely due to enhanced Marangoni flow and reduced surface tension effects.

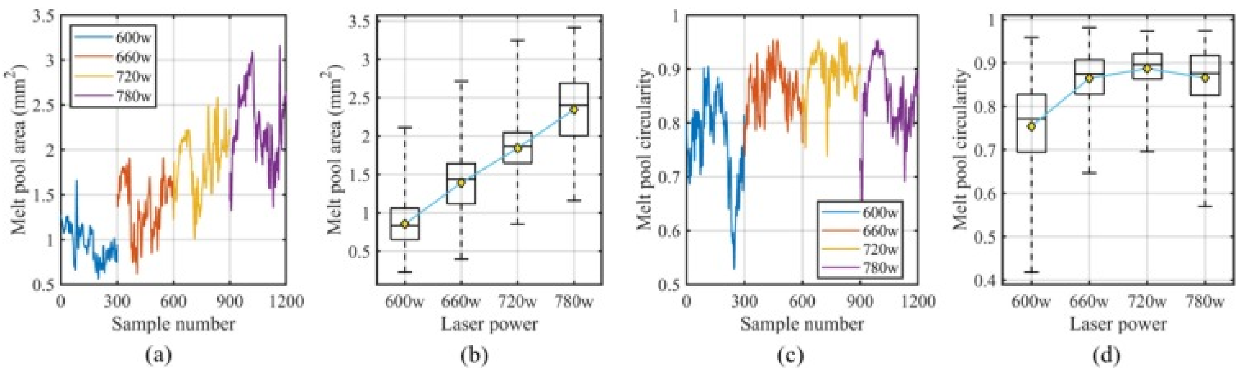

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Fig. 8. Melt pool characteristics at various laser powers. (a) Melt pool area vs. sample number. (b) Melt pool area distribution by laser power. (c) Melt pool circularity vs. sample number. (d) Melt pool irregularity distribution by laser power.

Following a quantitative analysis of melt pool geometric characteristics, examining the frequency-domain properties of photodetector signals can uncover nuanced dynamic variations inherent to the laser welding process. Morphological shifts in the melt pool often induce fluctuations across distinct frequency bands of these signals, thereby reflecting periodic oscillations and transient instabilities in laser-material interactions. As illustrated in Fig. 9, three signal processing techniques were systematically applied:

1. Fast Fourier Transform (FFT) transformed time-domain photoelectric signals into frequency spectra, pinpointing dominant frequency components and their harmonic signatures; The formula is as follows:

$$\mathrm{X}(k) = \sum_{n=0}^{N-1} \mathrm{X}[n] \cdot e^{-i\frac{2\pi}{N}kn}, k = 0, 1, \ldots, N-1 \tag{5}$$

where $X[n]$ represents the time-domain discrete sequence of the input signal, indicating the sampled values at time points n (where $n = 0, 1, \ldots N-1$). N denotes the total length of the signal. K is the frequency-domain index, corresponding to the discrete frequency points $f_k = \frac{k \cdot f_s}{N}$, where f_s is the sampling rate. $X(k)$ is the output frequency-domain complex sequence, which includes the magnitude $|X(k)|$ and phase $\arg(|X(k)|)$ at frequency f_k, reflecting the distribution of signal energy in the frequency domain.

2. Short-Time Fourier Transform (STFT) generated time–frequency maps to track spectral evolution during transient welding phases; The formula is as follows:

$$X(m, k) = \sum_{n=0}^{L-1} X[n + mH] \cdot W[n] \cdot e^{-i\frac{2\pi}{L}kn} \tag{6}$$

where $X[n + mH]$ represents the segment of the time-domain signal within the time frame m, starting at position mH (where H is the step size, and $m = 0, 1, \ldots, M-1$). $W[n]$ is the window function, and in this paper, a Hamming window is used to suppress spectral leakage of the truncated signal, with the window length being L (where $n = 0, 1, \ldots, L-1$). H is the sliding step size, which determines the overlap rate between adjacent time windows. $X(m, k)$ is the time–frequency matrix, where the row index m represents the time frame, and the column index k corresponds to the frequency $f_k = \frac{k \cdot f_s}{L}$ The matrix elements are complex numbers, with the magnitude representing energy and the phase representing time–frequency characteristics.

3. Power Spectral Density (PSD) visualization quantified energy distribution patterns across frequency bands, revealing process-related modulation

effects. The formula is as follows:

$$S\left(k\right)=\frac{1}{N\cdot f_{s}}\left|X(k)\right|^{2} \tag{7}$$

where $X(k)$ is the frequency-domain sequence output by the FFT, where N is the total length of the signal, and f_s is the sampling rate.

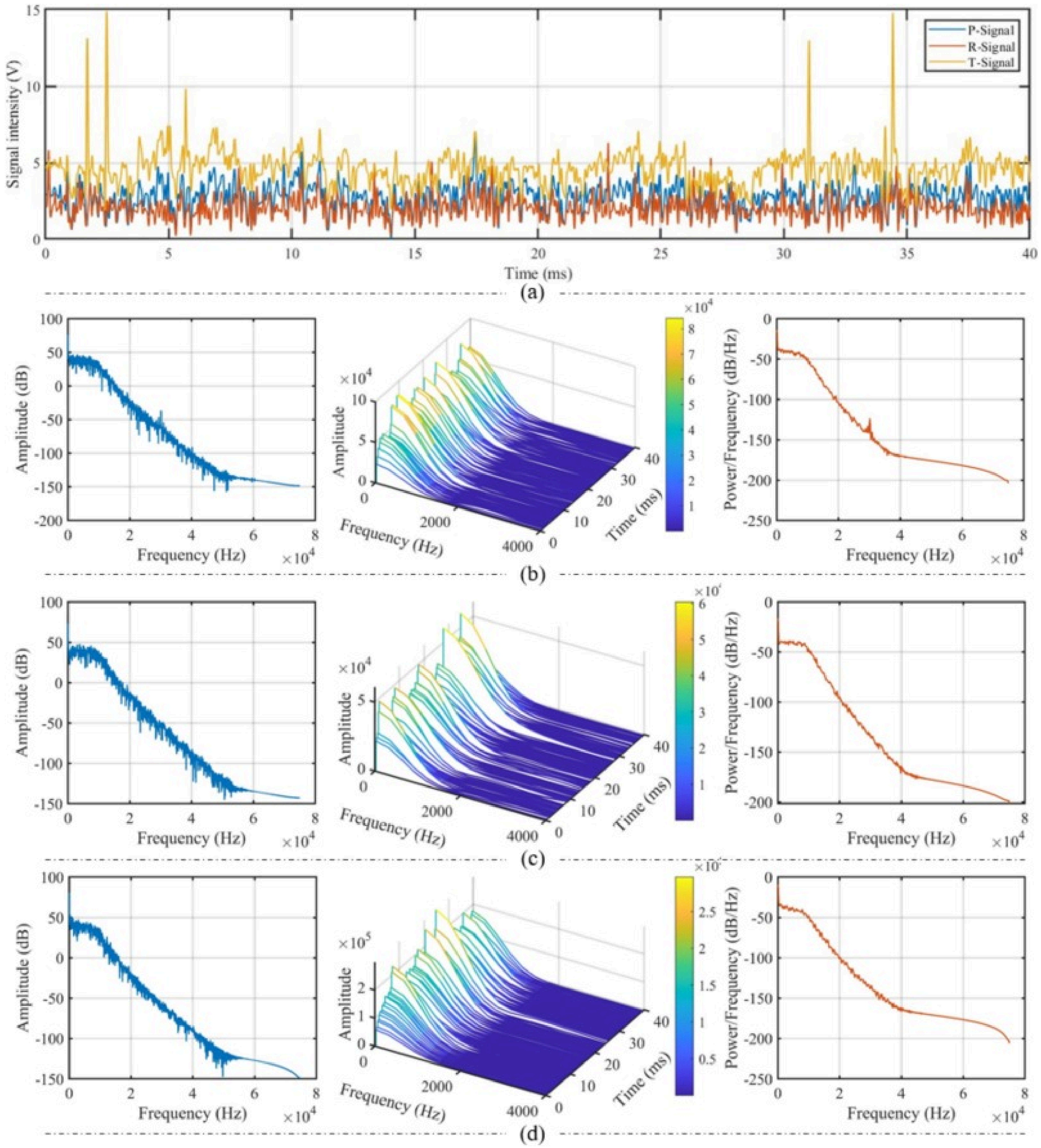

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Fig. 9. Characterization of optical radiation signals. (a) Raw signal. (b) Frequency domain characteristics of the P-Signal. (c) Frequency domain characteristics of the R-Signal. (d) Frequency domain characteristics of the T-Signal.

The photoelectric signal energy is predominantly concentrated in the low-frequency band (0–20 kHz), indicating that the macroscopic behavior of the melt pool serves as the primary information source. As depicted in Fig. 10, the melt pool spectrum exhibits dominant low-frequency components (<1 kHz) with stable spectral characteristics, suggesting that its dynamics are governed by low-frequency fluid flow and interfacial tension. In contrast, the keyhole spectrum displays significantly higher amplitude across broader frequency ranges, particularly showing pronounced fluctuations in the medium–low frequency regime (1–3 kHz), likely associated with transient metal vapor jet instabilities, collapse-rebound cycles, and rapid morphological reshaping within the keyhole. Consequently, the low-frequency components of optical radiation signals are strongly linked to melt pool oscillations, while medium–high frequency components correlate more closely with keyhole instability modes and vapor–liquid interaction dynamics. This observed correlation between low-frequency optical radiation components and melt pool dynamics can be effectively leveraged to enhance the performance of real-time weld quality monitoring systems. By focusing on the extraction and analysis of low-frequency signal features, the system can achieve improved robustness against high-frequency noise and transient disturbances. Furthermore, these low-frequency features, reflecting the thermal and morphological stability of the melt pool, can serve as reliable indicators of steady or unstable welding conditions.

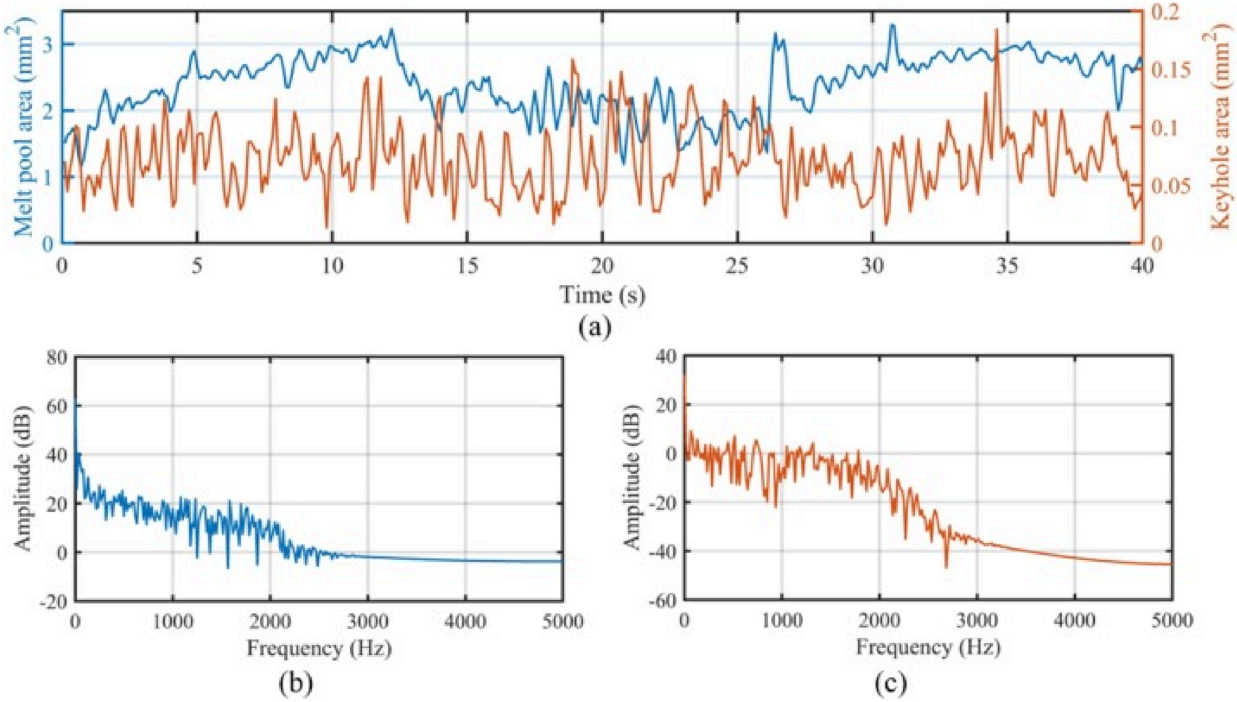

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Fig. 10. Melt pool and keyhole area dynamics and frequency domain characteristics. (a) Time variation of melt pool and keyhole area. (b) Frequency domain characteristics of melt pool. (c) Frequency domain characteristics of keyhole.

3.2. Impact of melt pool fluctuation on optical radiation signals

In laser welding of aluminum alloys, melt pool oscillation manifests as periodic fluctuations in signal intensity across three photoelectric sensors, corroborated by synchronized high-speed imaging. The infrared radiation emitted by the melt pool correlates directly with its temperature and surface area. Notably, the laser incidence zone exhibits elevated temperature and expanded surface area, generating intensified infrared radiation[30]. During oscillation, high-temperature liquid metal is displaced counter to the laser beam's travel direction, triggering measurable reductions in visible light radiation, infrared radiation, and laser reflection signals (Fig. 11). A stable melt pool (Fig. 11a) demonstrates synchronous attenuation of visible and infrared signals during minor backward oscillations. Concurrently, the keyhole undergoes rapid area contraction until momentary closure and reopening. The laser reflection signal exhibits heightened sensitivity to keyhole dynamics and melt pool morphology, displaying earlier and more pronounced transient changes compared to optical radiation signals. Due to the limited oscillation amplitude (<5% area variation), visible and infrared signal intensity deviations remain minimal, aligning with melt pool area trends. For enhanced visualization, dynamic regions in high-speed images are converted to heatmaps via frame differencing. Specifically, the absolute pixel-wise intensity differences between consecutive frames are computed to generate a difference image, which highlights regions of temporal variation. These pixel values are then normalized and mapped to a color scale, where higher intensity changes appear as warmer colors. The resulting heatmap intuitively reflects areas of significant melt pool and keyhole activity, such as oscillation, spatter, or collapse. Steady-state melt pool oscillations typically feature localized, low-amplitude perturbations (Fig. 11b). Under intensified oscillation conditions, the affected liquid metal volume expands spatially, concentrating near the keyhole periphery and melt pool boundaries. Crucially, the majority of molten material remains confined within the pool, ensuring negligible impact on weld bead formation.

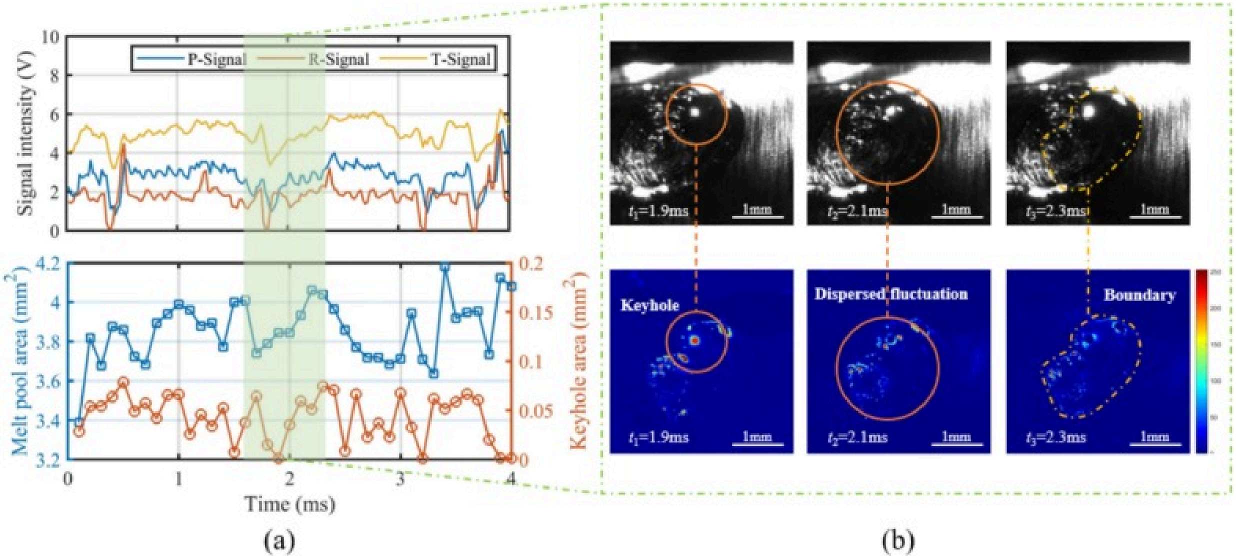

Fig. 11. The optical signals and related high-speed images in a typical slight oscillation situation. (a) Raw signals. (b) High-speed image at a slight oscillation point.

The most severe instability mode observed during aluminum alloy laser welding involves catastrophic ejection of the entire molten pool. This phenomenon triggers substantial metal loss that creates deep surface pits along weld seams, accompanied by exponential deterioration in both electrical conductivity and mechanical performance of joints [31]. Analysis of light radiation signatures reveals that such instabilities progress through a distinct developmental phase (Fig. 12a), during which measurable shifts in the melt pool's natural oscillation frequency precede the emergence of amplified surface fluctuations (Fig. 12b). These hydrodynamic perturbations induce localized turbulent flow in the molten metal, triggering erratic laser reflection signals marked by three transient intensity spikes within 0.5 ms. Concurrent surges in visible and infrared radiation intensities correlate with rapid temperature escalation in localized regions, driving explosive material vaporization and the formation of high-density metal vapor/plasma plumes that exacerbate optical signal instability through multi-photon absorption and scattering effects. Progressive laser energy deposition and internal pressure accumulation overcome equilibrium constraints in the melt pool, culminating in liquid metal ejection (2 ms) accompanied by spatter formation. This breakthrough-phase instability triggers abrupt termination of laser-liquid coupling and keyhole collapse. Post-ejection, rapid thermal dissipation reduces localized energy density, evidenced by precipitous drops in visible (300–700 nm) and infrared (1100–2100 nm) radiation intensities. Concurrently, surface tension-driven readjustments propagate radially from the detachment epicenter, stabilizing molten pool dynamics as reflected in recovering laser reflection signals. The subsequent energy reabsorption phase initiates temperature resurgence through renewed laser coupling, driving intensified metal vaporization and radiative emissions. Surface morphology alterations from violent hydrodynamic fluctuations modify optical scattering patterns, while pressure-surface tension hysteresis during spatter nucleation correlates directly with transient signal variations. Multi-spectral photodetector analysis reveals this self-limiting instability cycle through synchronized monitoring of thermal, optical, and hydrodynamic transients.

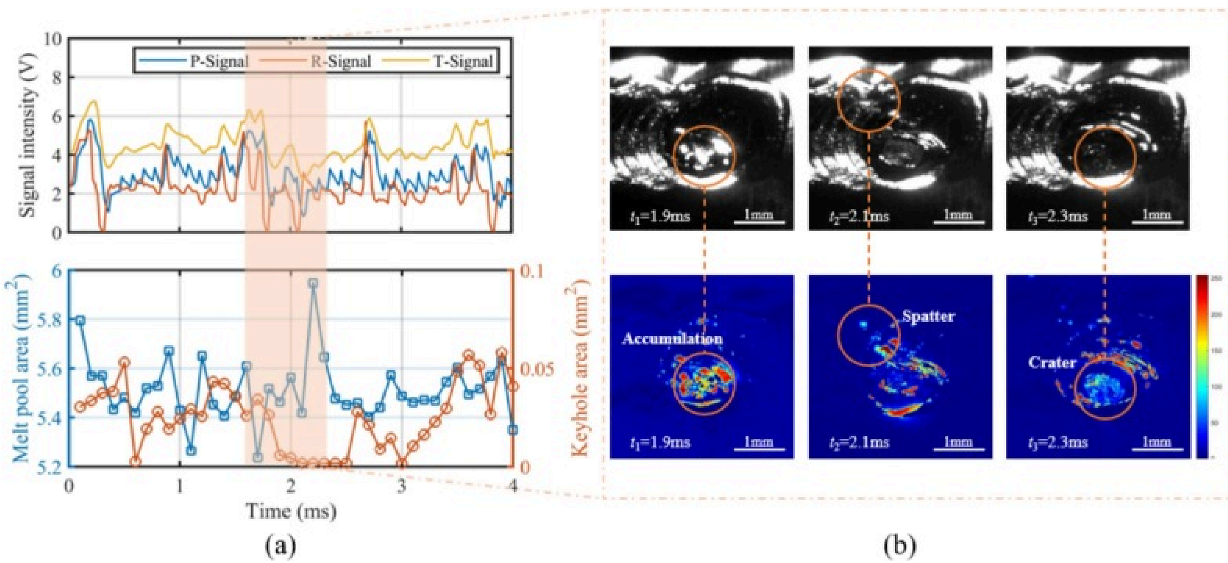

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Fig. 12. The optical signals and related high-speed images at a melt ejection case. (a) Raw signals. (b) Relative high-speed images in ejection process.

3.3. Association between signal characteristics and formation quality

The surface morphology and fusion bonding of the foil serve as critical indicators of welding quality in foil stack welding, directly influencing electrical conductivity [32,33]. Fig. 13 displays the weld's surface morphology and cross-sectional view. At 600W laser power (Fig. 13(a1)), the weld exhibits significant undulations, irregular defects, and rough texture with protrusions/depressions, indicating insufficient energy for complete material melting. This results in an unstable melt pool, rapid cooling, and subsequent surface roughness. The cross-section (Fig. 13(a2)) shows prominent defects and ineffective foil-backing bonding. At 660W power, the surface becomes smoother with reduced undulations and improved uniformity (Fig. 13(b1)), demonstrating enhanced welding quality through better color consistency. The cross-section at 660W power (Fig. 13(b2)) confirms incomplete penetration. At 720W, the weld achieves full flattening with superior smoothness (Ra < 10 μm) and minimal surface defects. The height differential (<49.97 μm) demonstrates exceptional uniformity, attributed to enhanced thermal input enabling complete melt pool fluidity. Cross-sectional analysis (Fig. 13(c1)) reveals complete metallurgical bonding between foil stack and backing piece. When power increases to 780W, the surface maintains integrity but develops periodic undulations (15–20 μm height differential), indicating excessive energy input. Fig. 13(d) demonstrates that penetration depth exceeding 0.9 mm compromises structural stability through over-penetration.

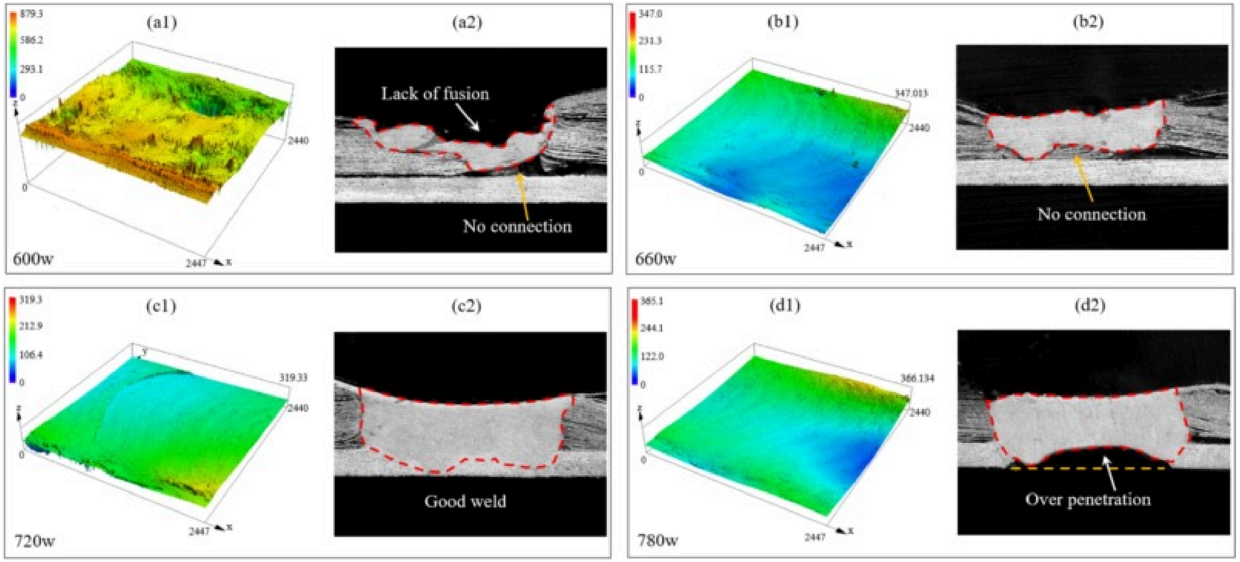

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Fig. 13. Comparative analysis of weld cross-section and surface topography under different laser powers.

To quantitatively analyze the surface morphology of the weld, two surface roughness characteristics, Arithmetic Mean Height (*Sa*) and Root Mean Square Height (*Sq*), were selected as evaluation metrics. *Sa* represents the arithmetic mean of the surface height deviations within a given area, and the calculation formula is as follows:

$$Sa = \frac{1}{A}\iint_A |z(x,y) - \overline{z}|dxdy \tag{8}$$

where, A represents the area of the evaluation region, $z(x,y)$ is the surface height at a specific point within the region, and $\overline{z}$ is the average height of all points within the region. This formula calculates the arithmetic mean of the surface height deviations within a given area. *Sq* represents the root mean square value of the surface height deviations within a given area, and is more sensitive to changes in surface roughness, particularly for larger peak-to-valley values. The calculation formula for *Sq* is as follows:

$$Sq = \sqrt{\frac{1}{A}\iint_A \left(z\left(x,y\right) - \overline{z}\right)^2 dxdy} \tag{9}$$

Sq emphasizes larger deviations by squaring the height deviations at each point, making it more sensitive to outliers in the surface texture. Fig. 14 illustrates the variations in weld surface roughness (*Sa* and *Sq*) and melt pool circularity (*Mc*) under different laser power levels. Laser power has a significant impact on both surface roughness and melt pool circularity. Within a certain range, increasing laser power reduces surface roughness and improves the circularity of the melt pool. However, when the laser power exceeds a certain threshold, further increases may lead to an increase in surface roughness, likely due to overheating or excessive melting of the melt pool caused by too high a laser power. Additionally, there is a negative correlation between roughness and circularity. This negative correlation suggests that a more circular melt pool shape is often associated with a smoother weld surface, possibly because the stability and uniformity of the melt pool shape directly influence the smoothness of the weld surface.

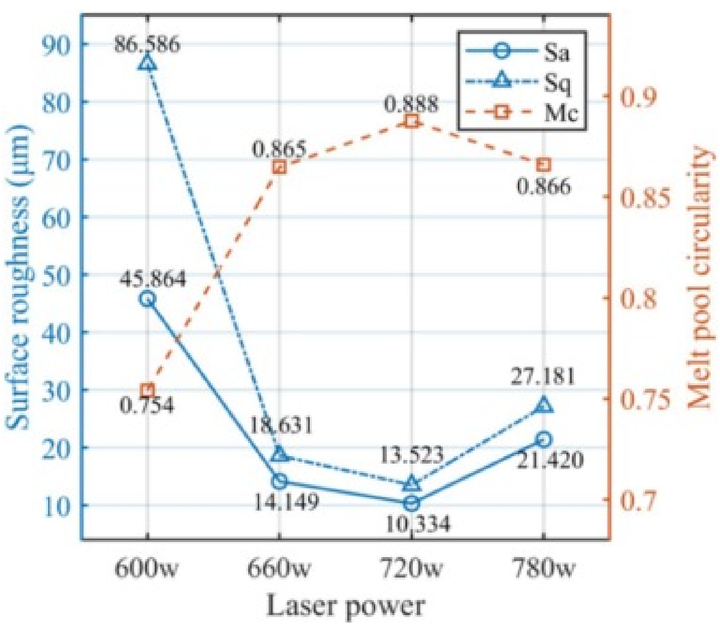

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Fig. 14. Variation of surface roughness and melt pool circularity with laser power.

To statistically investigate the intrinsic relationship between photoelectric signals and welding process characteristics, this section analyzes correlations among melt pool area, keyhole area, penetration depth, surface roughness, keyhole closure frequency, and photoelectric signals. Fig. 15 displays scatter plots with linear fittings of melt pool/keyhole areas versus photoelectric signals under different laser powers, revealing statistically significant correlations. In Fig. 15(a), a strong positive correlation exists between optical signal intensity and melt pool area, with signal magnitude increasing proportionally to area expansion. This trend remains consistent across laser power variations, though overlapping data points suggest laser power influences signal intensity through mechanisms beyond melt pool area alone, potentially involving keyhole dynamics or spatter effects. Fig. 15(b) demonstrates a weaker positive correlation for laser reflection signals versus melt pool area, evidenced by a reduced fitting slope compared to visible light signals. The reflection signal variability likely stems from melt pool surface conditions (e.g., roughness, oscillations), which are jointly governed by laser power and welding parameters. This explains the pronounced signal overlap observed at specific power levels (600W and 660W). Fig. 15(c) demonstrates a strong positive correlation between infrared radiation signals and melt pool area, paralleling visible light signal trends. The infrared signals display broader intensity ranges, suggesting enhanced sensitivity to melt pool temperature dynamics. Data dispersion increases across power conditions, particularly at higher power settings (720W and 780W), exhibiting amplified infrared responses. As shown in Fig. 15(d-e), all three signals maintain positive correlations with keyhole area, though correlation strength and distribution patterns differ.

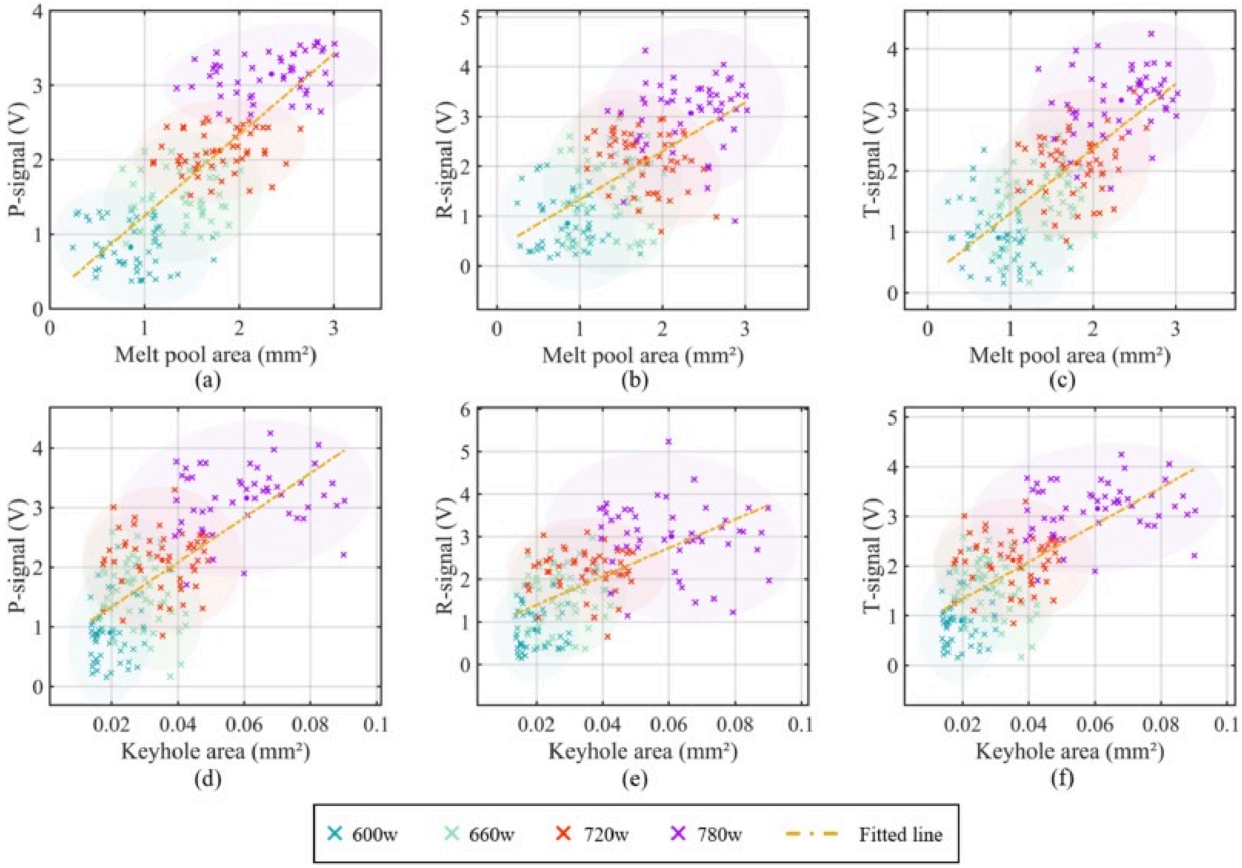

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Fig. 15. Correlation graphs between measured weld dimensions and signals for photodiodes.

The P-signal and T-signal are more sensitive to changes in keyhole area, while the R-signal appears to be more influenced by the surface reflection characteristics of the keyhole. Power has a significant impact on the signals, especially at higher power conditions, where the data points exhibit a wider distribution range. This suggests that power not only alters the physical characteristics of the keyhole but also affects the intensity and stability of the photoelectric signals. For melt pool area, the fitting lines for all three signals have relatively shallow slopes, indicating that the changes in signal intensity mainly reflect the gradual accumulation of melt pool expansion. In contrast, for keyhole area, the fitting lines have steeper slopes, particularly for the T-signal, suggesting that keyhole area changes have a more direct and pronounced effect on signal intensity. This could be related to the fact that melt pool area is predominantly governed by heat conduction, while keyhole area is more directly influenced by laser energy coupling intensity and material evaporation dynamics. Overall, the P-signal primarily reflects the overall optical scattering effects during welding, correlating with the total area changes of both the keyhole and melt pool. The overlap in data distribution and the lower slope of the fitting line suggest that the R-signal is more susceptible to power fluctuations and changes in surface reflection angles, and its physical mechanism requires further investigation. The T-signal shows the most significant positive correlation in both area changes, particularly with keyhole area at higher power conditions, indicating that the T-signal is highly responsive to variations in thermal radiation intensity, which predominantly originates from the high-temperature region around the keyhole and its dynamic changes.

By analyzing the correlation between quantitative features and photoelectric signals, the intrinsic relationship between signal characteristics and the dynamic behavior of the melt pool and keyhole during the welding process can be revealed. As shown in Fig. 16, both melt pool area (Ma) and penetration depth (Pd) exhibit a significant positive correlation with the mean values of photoelectric signals, with the T-signal mean (Ut) showing the strongest correlation, reaching 0.87 and 0.79, respectively. This suggests that the T-signal mean is most sensitive to changes in melt pool size and penetration depth, making it suitable for monitoring these parameters. Additionally, the means of the P-signal and R-signal also exhibit a strong positive correlation with melt pool area and penetration depth, with correlation coefficients ranging from 0.77 to 0.84.

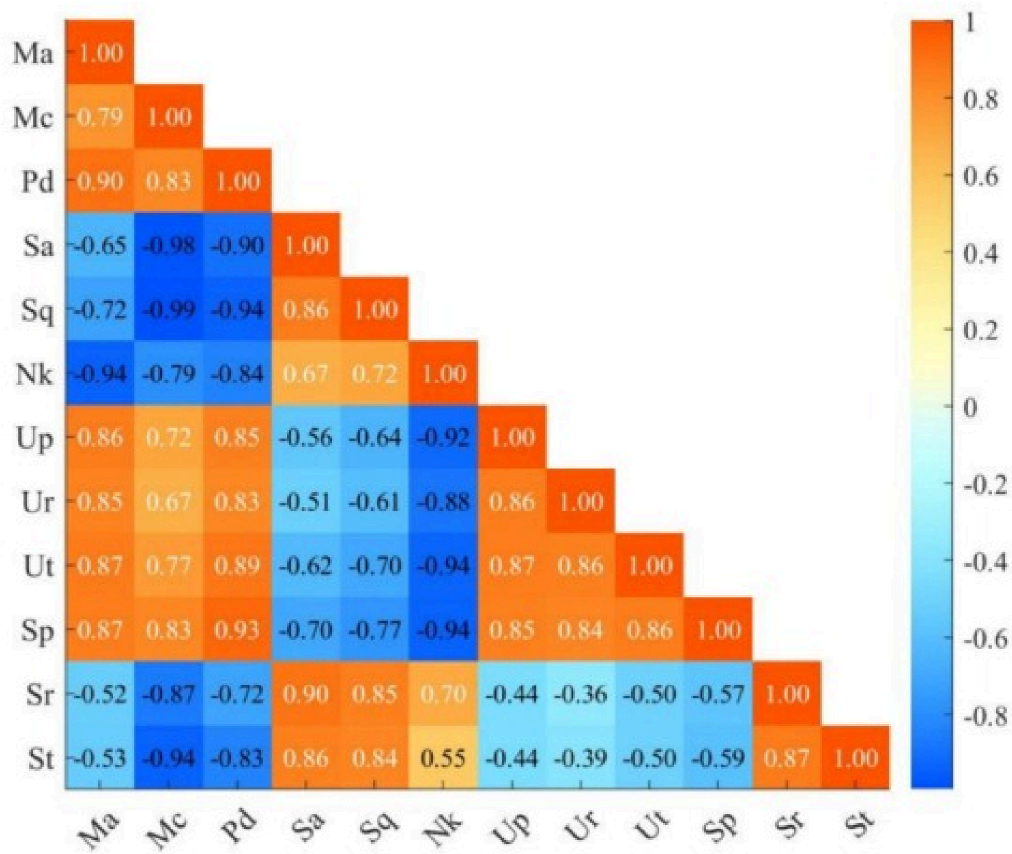

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Fig. 16. Correlation matrix between weld features (*Ma*, *Mc*, *Pd*, *Sa*, *Sq*, and *Nk*) and signals features (*Up*, *Ur*, *Ut*, *Sp*, *Sr* and *St*).

On the other hand, surface roughness (*Sa*) is significantly negatively correlated with the mean values of photoelectric signals (correlation coefficient ranging from −0.58 to −0.59), indicating that higher roughness may reduce signal strength, which is consistent with the impact of rough surfaces on light reflection and scattering. Therefore, reducing surface roughness can help improve signal stability and monitoring accuracy.

The standard deviation of photoelectric signals is also correlated with the dynamic characteristics of the melt pool and keyhole. Specifically, the correlation coefficient between *Ma* and the T-signal standard deviation (*St*) is 0.57, indicating that signal fluctuations significantly increase as the melt pool undergoes dynamic changes. Additionally, the correlation coefficient between keyhole closure frequency (*Nk*) and the T-signal standard deviation is 0.52, suggesting that signal fluctuations can be used to characterize the intensity and frequency of keyhole dynamic behavior. However, the standard deviation of the R-signal shows weaker correlations with other quantitative features, indicating its limited ability to reflect melt pool and keyhole dynamics.

The T-signal demonstrates a good response to welding process dynamics in both mean value and variability dimensions, particularly with strong sensitivity to changes in melt pool area and keyhole dynamics. In the future, a multi-signal fusion strategy could be employed, using the P-signal and R-signal for melt pool boundary monitoring, while the T-signal could be the primary signal for monitoring thermal radiation and keyhole behavior. Furthermore, utilizing dynamic time modeling or machine learning methods to further explore the nonlinear relationships between signal features and welding dynamics may enhance the accuracy and robustness of monitoring methods.

4. Conclusion

In this work, a new perspective on the photoelectric process monitoring mechanism is proposed by establishing the relationship between multidimensional light radiation signals and keyhole/melt pool characteristics. This provides solid data support for understanding the intrinsic connection between melt pool dynamic changes and welding process stability. The main findings are as follows:

(i) During the laser welding process, the variation in light radiation signals is closely related to the dynamics of the melt pool. Experimental results show that the low-frequency fluctuations of the melt pool are strongly correlated with the low-frequency components (0–2000 Hz) of the light radiation signal. The light radiation signal not only reveals the macroscopic motion and periodic changes of the melt pool but also provides a powerful quantitative basis for real-time monitoring of welding process stability.

(ii) The correlation between light radiation signals and instantaneous changes in the melt pool has been elucidated. By analyzing dynamic heat maps of the melt pool, its dynamic behavior is categorized into slight fluctuations and extreme instability states. A joint analysis of photoelectric signals and instantaneous features of the keyhole/melt pool is conducted. High-speed photography reveals that instantaneous fluctuations in photoelectric signals are directly related to melt pool oscillations and keyhole dynamic states.

(iii) The characteristics of photoelectric signals are closely related to weld formation quality. The mean values of optical signals (P-signal, R-signal, T-signal) exhibit significant positive correlations with melt pool area and penetration depth, while surface roughness is significantly negatively correlated with signal strength. The T-signal is particularly sensitive to changes in melt pool area and penetration depth, making it suitable for monitoring these features. These findings indicate that optical signals can effectively reflect the quality of weld formation and can be utilized to optimize the welding process.

In future work, it is essential to consider potential external factors in industrial environments—such as ambient light, surface contamination, mechanical vibrations, and material variability—that may affect the stability of optical signals. To improve robustness, future systems should integrate noise-resistant feature extraction, sensor fusion, and adaptive algorithms capable of handling environmental fluctuations and diverse welding conditions. Additionally, future investigations should prioritize the integration of multimodal signal fusion and intelligent modeling to establish robust multi-physics-coupled deep learning frameworks. By synergizing photoelectric signals with thermal–mechanical field data and acoustic emission signatures, advanced models could be developed to enhance real-time prediction accuracy and noise immunity for welding state monitoring. A promising avenue lies in constructing graph neural network (GNN)-based architectures to dynamically extract spatiotemporal features from heterogeneous sensor streams, explicitly decoding the nonlinear mapping between transient melt pool oscillations, keyhole collapse dynamics, and their multisource signal manifestations. This approach would enable precise quantification of process anomalies while providing mechanistic insights into the cross-physical interactions governing laser-material interactions. To further enhance prediction robustness under complex industrial conditions, additional sensor inputs—such as acoustic emission signals for detecting spatter and keyhole instability, and thermal imaging for capturing melt pool temperature distribution—can be incorporated. These complementary modalities would enrich the feature space and improve noise resilience in dynamic welding environments. Overall, this approach would enable precise, real-time quantification of weld quality and provide deeper mechanistic insights into the nonlinear laser–material interaction process.

CRediT authorship contribution statement

Da Zeng: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Di Wu:** Writing – review & editing, Validation, Software, Resources, Methodology, Funding acquisition, Conceptualization. **Peilei Zhang:** Writing – review & editing, Validation, Methodology, Funding acquisition. **Haichuan Shi:** Writing – review & editing, Validation, Supervision, Software. **Qinghua Lu:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology. **Zhishui Yu:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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